Osman Melih Cabik

Saarbrücken, Germany • M.Sc. Candidate in Educational Technology | Human-Computer Interaction & XR Systems

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ABOUT ME

M.Sc. student and Student Programming Tutor at Saarland University specializing in Human-Computer Interaction (HCI), Spatial Computing, and Interactive Simulation. Experienced in engineering real-time 3D interaction frameworks, Web/Unity telemetry dashboards, and simulation workflows in Unity 3D (Meta XR / OpenXR) and Unreal Engine 5 (C++). Combines robust software engineering with empirical cognitive evaluation (Cognitive Load Theory, NASA-TLX, user studies, interaction telemetry). Passionate about contributing to cutting-edge research in spatial AI, XR agent interaction, and expert decision-support systems as a Master's Thesis Researcher and Student Research Assistant (HiWi).

EDUCATION

Universität des Saarlandes **Saarbrücken, Germany**

M.Sc. Educational Technology — Current GPA: 1.7 / 1.0 *Oct 2025 – Present*

- **Core Focus:** Human-Computer Interaction (HCI), Spatial Computing & XR Systems, Cognitive Ergonomics & Load Theory, Empirical Research Methods & Usability Testing.

Istanbul University-Cerrahpasa **Istanbul, Türkiye**

Bachelor of Education in Preschool Teaching — Final Grade: 3.64 / 4.0 (~1.36 German Grade) *Sep 2018 – Jun 2022*

- **Core Focus:** Cognitive Load Theory (Sweller), Instructional Scaffolding, Human-Centered Empirical Evaluation.

EXPERIENCE & ACADEMIC PROJECTS

Universität des Saarlandes **Saarbrücken, Germany**

Student Teaching Assistant / Tutor — Programming for Educational Technology *Oct 2026 – Present*

- Lead weekly interactive programming tutorials and lab walkthroughs in **Python**, mentoring Master's students in fundamental programming paradigms, algorithmic thinking, and adaptive learning architectures.
- Conduct structured code reviews, diagnose runtime/syntax bugs, and guide students through systematic debugging and scaffolded exercises (worked examples, Parsons problems) to minimize extraneous cognitive load.

EdTech Campus Orientation Scavenger Hunt App **Saarbrücken, Germany**

Lead Full-Stack Web Developer & Orientation Organizer *Sep 2026 – Oct 2026*

- Architected and deployed a mobile-web campus orientation platform using **React 18**, **TypeScript**, and **Leaflet** (OpenStreetMap) utilized by incoming Master's cohorts during official faculty orientation.
- Implemented GPS geolocation range detection, client-side camera QR scanning via HTML5 MediaDevices API, dynamic checkpoint clue routing, and live leaderboard telemetry via **Supabase** (PostgreSQL).

Antigravity 2.0: Location-Based WebAR Simulation Platform **Saarbrücken, Germany**

Lead AR Developer (Dr. Kristin Altmeyer's XR Course • Showcased at XR Education Day) *Apr 2026 – Jul 2026*

- Architected a full-stack gamified WebAR outdoor simulation featuring real-time camera overlays, diegetic HUDs, frequency-tuner bypass mechanics, and GPS geofencing using **React 18**, **Leaflet**, **Supabase**, and **MediaDevices API**.
- Conducted empirical study design assessing intrinsic motivation (IMI), usability (SUS), and cognitive load (NASA-TLX); demonstrated live system mechanics to DFKI and university researchers during the XR Education Day.

Land of Chasers **Ankara, Türkiye**

Founder & Simulation Software Engineer (C++ & Python) *Oct 2022 – Jan 2026*

- Engineered interactive 3D simulation mechanics using **C++**, **Unreal Engine 5**, and **Unity 3D**, optimizing real-time rendering pipelines, execution speed, and user interaction responsiveness.
- Conceptualized and executed controlled test scenarios evaluating user interaction telemetry and applied HCI principles to minimize cognitive load during real-time 3D control.

Google - AI and Tech Academy **Ankara, Türkiye**

Product Owner Trainee & Cross-Platform Developer *Nov 2022 – Jul 2023*

- Co-developed EduCon mobile application (**Flutter**, **Firebase**), conducting Agile/Scrum sprint planning, API integrations, and UI/UX usability reviews.

SKILLS & LANGUAGES

XR & Spatial Computing: Unity 3D (Meta XR SDK, OpenXR, Hand Tracking), Unreal Engine 5 (C++ API, Blueprints), WebXR, Spatial Telemetry & Interaction Logging.

Programming & Web Architectures: Python, C++, C#, TypeScript, JavaScript, React 18, Flutter, HTML5/CSS3, REST APIs.

Empirical Research & Evaluation: Cognitive Load Theory (CLT), NASA-TLX, Usability Testing (SUS), User Study Design, Interaction Telemetry Analysis.

DevOps & Infrastructure: Git & GitHub Actions, Supabase (PostgreSQL), Firebase, CMake / Build Toolchains, Agile & Scrum.

Languages: English (C1 - Fluent / Professional), German (B1 - Intermediate / Enrolled in B2 Course), Turkish (Native).